



Identifying Interdisciplinary Challenges Across 3D Data Creation/Curation Workflows

M McCall Borozan¹, Zack Lischer-Katz¹ and Matt Cook²

¹ College of Information Science, University of Arizona, United States

² Harvard College Library, Harvard University, United States

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Abstract

This study presents findings from an in-progress grant-funded project using a multi-case study design to explore the use of 3D and VR in academic libraries and research contexts in the United States. In a 2019 forum on 3D/VR Technologies in the U.S., participants recommended developing generic 3D data creation/curation workflows (Cook et al., 2019). Published 3D data curation standards often describe abstract protocols, but applying generic workflows is imprecise given the variety of 3D data (Altenhöner et al., 2023; Cook, 2024; Denard, 2012; Grayburn et al., 2019). Project-specific workflows require significant development for different use-cases, so here we synthesize “naturally occurring” 3D data workflows derived from 3D practitioners for cross-discipline comparison of 3D creation/curation practices (Rourk, 2019; Smith, 2008) and an evaluation of their respective conformity to FAIR data principles. Empirically derived 3D workflows reveal common challenges and areas of potential institutional support (Post & Chassanoff, 2021) to enhance the “FAIRness” of 3D data pipelines.

To address the question: ‘How can institutions and curation professionals best support 3D data creation, analysis, and curation?’, qualitative data were collected from ~20 labs, groups of practitioners regularly producing 3D content, across 6 U.S. institutions, in a range of disciplines from the natural sciences to the humanities. Semi-structured interviews and workflow observations of 47 participants were collected from 2024-2025, focused on participants’ roles in the processes of 3D content creation and curation. Transcripts are being analysed using qualitative-interpretive coding for 3D data pipelines, capturing the stages, training, and tools involved.

An experimental facet of this project concurrently tested the potential for immersive tools (XR) to support 3D data workflows. Participants engaged with generalizable 3D quality control tasks in virtual reality and on a tablet computer (i.e., control), and their relative performance was evaluated (Cook et al., 2026). Project development is member-checked and aligned at community advisory board meetings twice a year.

Preliminary results show that while lab workflows vary across and within disciplines, practitioners share workflow development practices and curation challenges. Interpretations of *final* 3D output are often based on a practitioner’s judgement,

determined by lab research questions. Practitioners in cultural heritage preservation and plant sciences identified *data transfer to storage* as a bottleneck; institutional storage is typically unavailable or too small to host 3D data. Metadata and paradata practices draw on a blend of domain-specific and locally developed guidelines, with support provided by lab members rather than reference to a standard. These findings align with the literature documenting *ad hoc* 3D workflow solutions developed to suit individual data goals, and the technical craft inherent to data curation (Cook et al., 2019; Grayburn et al., 2019; Thomer et al., 2022). Institutions could provide data storage and technical librarians familiar with 3D techniques to support these practices. XR experimental results offer the potential for XR to address points of breakdown in *Process* and *Post-Process* stages of 3D data workflows, as XR was found to reduce time and perceived effort on 3D data quality control tasks (Cook et al., 2026).

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